SERVICE MANUAL

USA Instruments, Inc.

Document 2413107-3

Revision 2

GE Signa® EXCITE™ 3.0T
Premier III Phased Array CTL Spine Coil
(3.0T HD CTL Array)

GE Catalog Part Number: M1385AW

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Rev. 08/15/2003

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- Si proceda alla manutenzione dell’apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
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SECTION 1 – INTRODUCTION

1-1 Product Identification and Shipping List

To identify the Premier III Phased Array CTL Spine Coil, refer to the rating plate for GE part number 2413107. *Figure 1* shows a picture of the 3T CTL Coil.

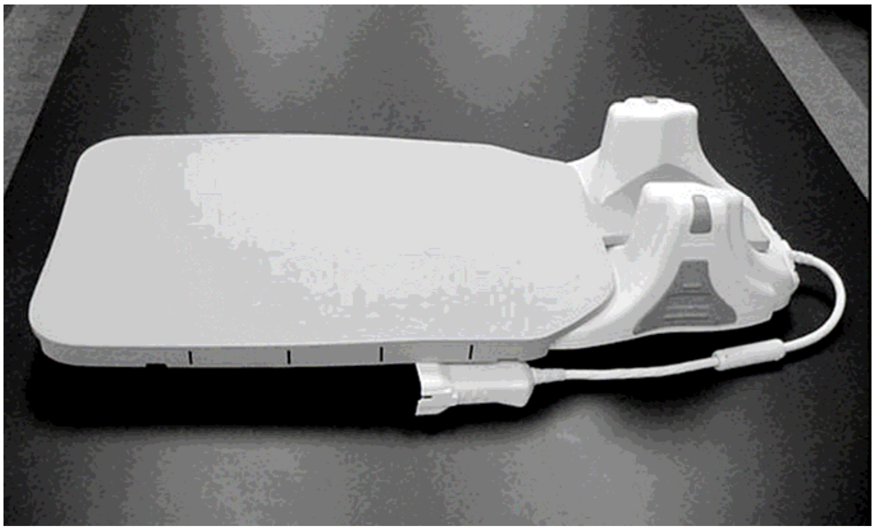

Figure 1: 3T CTL Coil

SHIPPING LIST – TABLE 1-1

Box #	Part Name	GE Part #	Qty
1	Premier III Phased Array CTL Spine Coil w/ Cable	2413107	1
1	Patient Comfort Pad	2413401	1
1	Ear Pads	2414423	2
1	Disposable Ear Pad Covers	U1-150536	500
1	Operator Guide Packet	2422232-500	1
1	Service Manual	2413107-3	1
2	3T 8Ch CTL Phantom CT Section	2381682-9	1
3	3T 8Ch CTL Phantom TL Section	2381682-10	1
4	3T 8Ch CTL Phantom TL Section	2381682-10	1

1-2 Compatibility

This coil is compatible with the Signa® Excite 3.0T system.

1-3 Related Documentation

Operator's Guide, GE Part Number 2422232.

Signa® EXCITE 3T 11.x Service Methods CD, 2389160-1.

Signa® EXCITE 3T 12.x Service Methods CD, 5133330-1.

1-4 Environmental Requirements

Store the coil in an air-conditioned scan room or equipment room.

Environmental Requirements – TABLE 1-4

	Transport and Storage	Operating Conditions
Barometric Pressure	645hPa to 795hPa	
Humidity	20% to 95% non-condensing	45% to 75% non-condensing
Temperature	-15° C to 55° C (5° F to 131° F)	21° C to 35° C (69.8° F to 95° F)

1-5 Theory of Operation

The Premier III Phased Array CTL Spine Coil is a receive-only coil with a total of 12 coil elements, 6 loop elements and 6 saddle elements (see *Figure 2*). It is designed to give optimum signal to noise ratio and uniform coverage of the spine anatomy, including the cervical, thoracic and lumbar regions. It can be operated in 12 different modes. An electronic multiplexer inside the coil, controlled by the MR system, selects the coil elements for each mode (see *Figure 3*). The unselected coil elements are turned off by the DC currents provided by the MR system. *Table 1-5* shows the selected elements and clinic use of each mode.

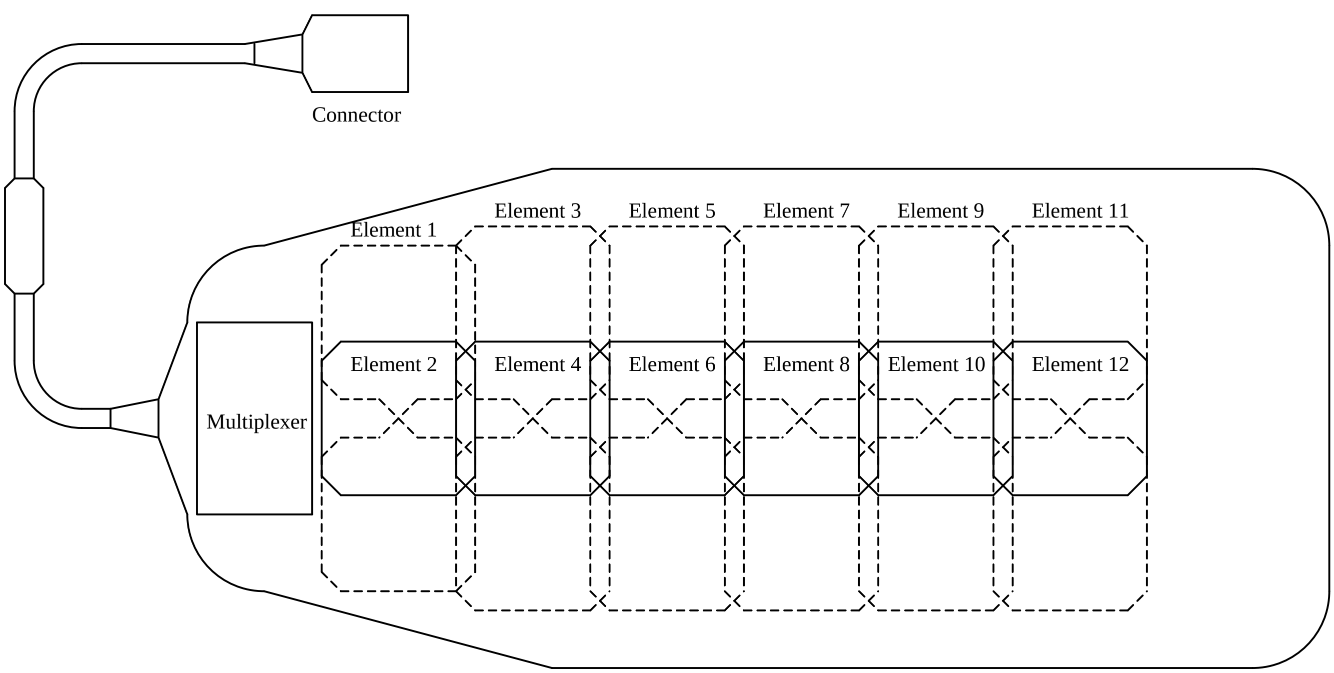

Figure 2: The layout of coil elements

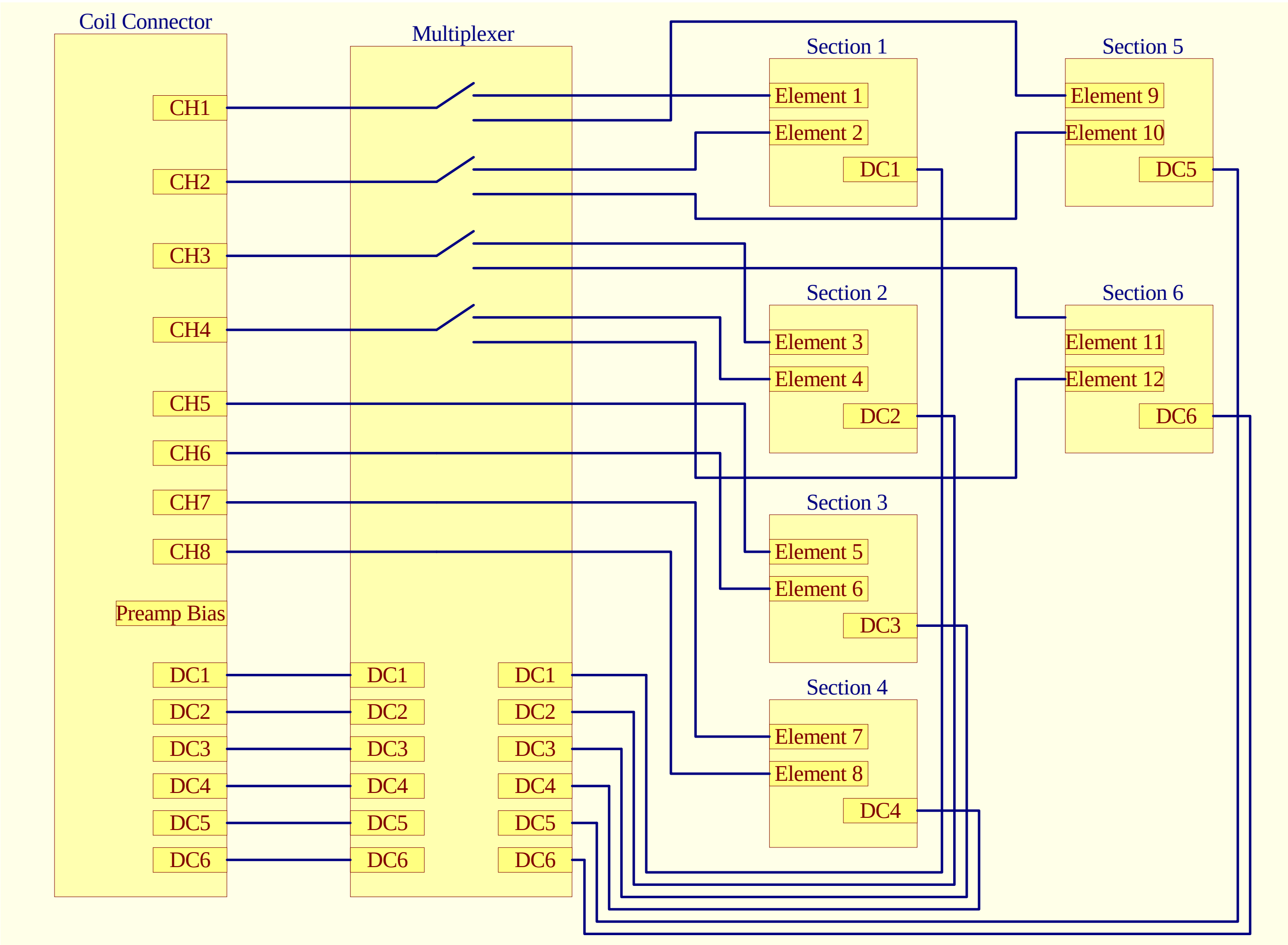

Figure 3: Coil Diagram

Modes of Operation – TABLE 1-5

Coil Mode (for 11.0 system)	Coil Mode (for 12.0 system)	Selected Elements	Clinical Use
8USCTLTOP	HDCTL TOP	1, 2, 3, 4, 5, 6, 7, 8	Cervical and thoracic spine
8USCTLMID	HDCTL MID	3, 4, 5, 6, 7, 8, 9, 10	Thoracic and lumbar spine
8USCTLBOT	HDCTL BOT	5, 6, 7, 8, 9, 10, 11, 12	Thoracic and lumbar spine
8USCTL123	HDCTL 123	1, 2, 3, 4, 5, 6	Cervical and thoracic spine
8USCTL234	HDCTL 234	3, 4, 5, 6, 7, 8	Thoracic spine
8USCTL345	HDCTL 345	5, 6, 7, 8, 9, 10	Thoracic and lumbar spine
8USCTL456	HDCTL 456	7, 8, 9, 10, 11, 12	Thoracic and lumbar spine
8USCTL12	HDCTL 12	1, 2, 3, 4	Cervical spine
8USCTL23	HDCTL 23	3, 4, 5, 6	Thoracic spine
8USCTL34	HDCTL 34	5, 6, 7, 8	Thoracic spine
8USCTL45	HDCTL 45	7, 8, 9, 10	Thoracic and lumbar spine
8USCTL56	HDCTL 56	9, 10, 11, 12	Lumbar spine

SECTION 2 – SETUP AND CALIBRATION

2-1 Coil Installation

2-1-1 Special Install Notes

None.

2-1-2 Configuration

This coil uses the following configuration names in 11.0 system:

8USCTLTOP
8USCTL MID
8USCTLBOT
8USCTL123
8USCTL234
8USCTL345
8USCTL456
8USCTL12
8USCTL23
8USCTL34
8USCTL45
8USCTL56
8USCTL123service
8USCTL456service
8USCTL12service
8USCTL34service
8USCTL56service

This coil uses the following configuration names in 12.0 system:

HDCTL TOP
HDCTL MID
HDCTL BOT
HDCTL 123
HDCTL 234
HDCTL 345
HDCTL 456
HDCTL 12
HDCTL 23
HDCTL 34
HDCTL 45
HDCTL 56
HDCTL 123serv
HDCTL 456serv
HDCTL 12serv
HDCTL 34serv
HDCTL 56serv

Add the coil configuration names using the Configuration File Manager. Refer to: Service Methods CD; Software; System Level Software Utilities; Configuration File Manager.

The following tables provide additional information about the coil. *Table 2-1-1* shows the relation between coil modes, receive channels and selected coil elements. *Table 2-1-2* shows the relation between coil modes and MC bias.

Coil modes, Receive channels and selected coil elements – Table 2-1-1

Coil Mode		Receive Channel							
Coil Mode for 11.0 System	Coil Mode for 12.0 System	CH 1	CH 2	CH 3	CH 4	CH 5	CH 6	CH 7	CH 8
8USCTLTOP	HDCTL TOP	E1	E2	E3	E4	E5	E6	E7	E8
8USCTLMID	HDCTL MID	E9	E10	E3	E4	E5	E6	E7	E8
8USCTLBOT	HDCTL BOT	E9	E10	E11	E12	E5	E6	E7	E8
8USCTL123	HDCTL 123	E1	E2	E3	E4	E5	E6		
8USCTL234	HDCTL 234			E3	E4	E5	E6	E7	E8
8USCTL345	HDCTL 345	E9	E10			E5	E6	E7	E8
8USCTL456	HDCTL 456	E9	E10	E11	E12			E7	E8
8USCTL12	HDCTL 12	E1	E2	E3	E4				
8USCTL23	HDCTL 23			E3	E4	E5	E6		
8USCTL34	HDCTL 34					E5	E6	E7	E8
8USCTL45	HDCTL 45	E9	E10					E7	E8
8USCTL56	HDCTL 56	E9	E10	E11	E12				
8USCTL123 service	HDCTL 123serv	E1	E2	E3	E4	E5	E6		
8USCTL456 service	HDCTL 456serv	E9	E10	E11	E12			E7	E8
8USCTL12 service	HDCTL 12serv	E1	E2	E3	E4				
8USCTL34 service	HDCTL 34serv					E5	E6	E7	E8
8USCTL56 service	HDCTL 56serv	E9	E10	E11	E12				

Coil modes and MC bias – Table 2-1-2

Coil Mode		MC Bias							
Coil Mode for 11.0 System	**Coil Mode for 12.0 System**	**MC1**	**MC2**	**MC3**	**MC4**	**MC5**	**MC6**	**MC7**	**MC8**
8USCTLTOP	**HDCTL TOP**	Toggle	Toggle	Toggle	Toggle			Not used	Not used
8USCTLMID	**HDCTL MID**		Toggle	Toggle	Toggle	Toggle		Not used	Not used
8USCTLBOT	**HDCTL BOT**			Toggle	Toggle	Toggle	Toggle	Not used	Not used
8USCTL123	**HDCTL 123**	Toggle	Toggle	Toggle				Not used	Not used
8USCTL234	**HDCTL 234**		Toggle	Toggle	Toggle			Not used	Not used
8USCTL345	**HDCTL 345**			Toggle	Toggle	Toggle		Not used	Not used
8USCTL456	**HDCTL 456**				Toggle	Toggle	Toggle	Not used	Not used
8USCTL12	**HDCTL 12**	Toggle	Toggle					Not used	Not used
8USCTL23	**HDCTL 23**		Toggle	Toggle				Not used	Not used
8USCTL34	**HDCTL 34**			Toggle	Toggle			Not used	Not used
8USCTL45	**HDCTL 45**				Toggle	Toggle		Not used	Not used
8USCTL56	**HDCTL 56**					Toggle	Toggle	Not used	Not used
8USCTL123 service	**HDCTL 123serv**	Toggle	Toggle	Toggle				Not used	Not used
8USCTL456 service	**HDCTL 456serv**				Toggle	Toggle	Toggle	Not used	Not used
8USCTL12 service	**HDCTL 12serv**	Toggle	Toggle					Not used	Not used
8USCTL34 service	**HDCTL 34serv**			Toggle	Toggle			Not used	Not used
8USCTL56 service	**HDCTL 56serv**					Toggle	Toggle	Not used	Not used

The CTL coil use +7V for forward bias. The blank blocks in the above table means the MC bias is un-toggled.

2-2 Installation Functional Checks

Perform Section 3-2 Coil Imaging Performance Verification.

2-3 Periodic Quality Assurance Check

On a periodic basis, such as during planned maintenance, perform the quality assurance checks as outlined below to ensure the coil is operating properly.

1. Check external cable for cracks or cuts.
2. Perform Section 3-2 Coil Imaging Performance Verification.

SECTION 3 – FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TOOLS REQUIRED – TABLE 3-1

Description	GE Part #	Qty
CT Phantom	2381682-9	1
TL Phantom	2381682-10	2

3-2-2 Data Collection

Step 1 – Place the CTL coil on the patient cradle. Note the serial number of the coil.

Step 2 – Remove patient comfort pad and position the phantoms on the coil as shown in *Figure 4*.

Step 3 – Place the Landmark on the raised bar on the side of the coil as shown in see *Figure 5*.

Figure 4: Positioning the coil and phantoms

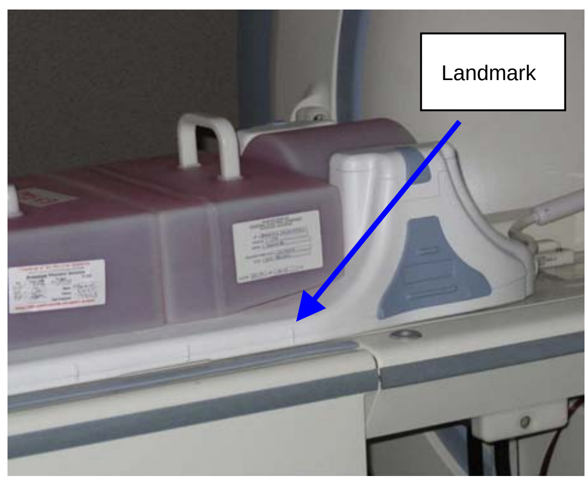

Figure 5: Landmark

Step 4 – Move back to the operator workspace and end the previous exam (if required.)

Step 5 – For 12.x or earlier systems: From the Common Service Desktop, select **Image Quality**, and then **Coil SNR**. Then select **Click here to start this tool**.

For 14.x or later systems: From the Common Service Desktop, select **Image Quality**, and then **MCQA Tool**. Then select **Click here to start this tool**. For additional information regarding MCQA, refer to the 14.x or later Service CD at the following location: troubleshooting/Image Quality/Multicoil Quality Assurance.

Step 6 – Select the [**8- Channel CTL Array**] from the Coil Name selection menu on the Multicoil SNR Tool Control window (see *Figure 6*).

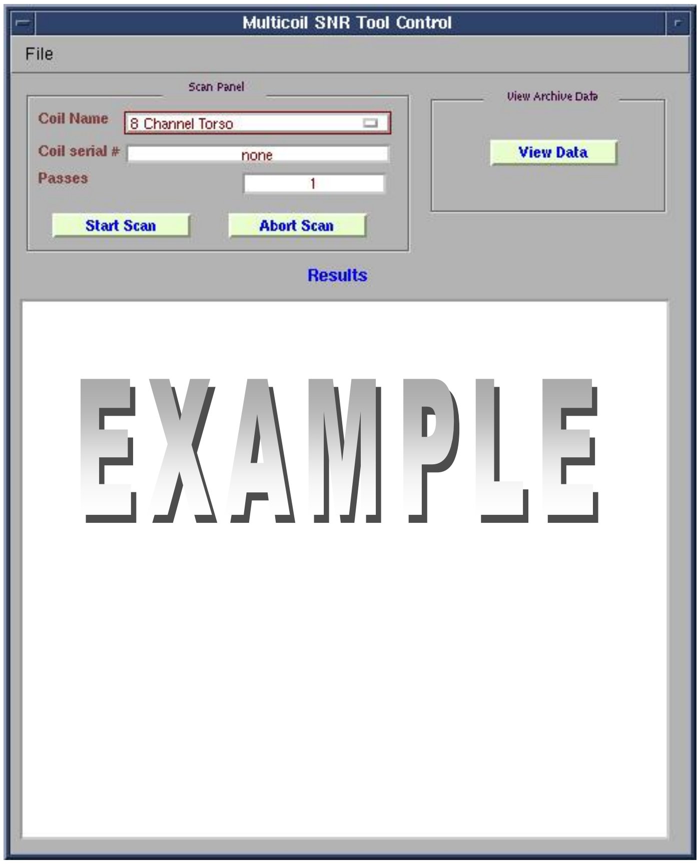

Figure 6: Multicoil SNR Tool Control window.

Step 7 – Enter the Coil serial number.

Step 8 – Enter the number of repetitions in the Passes field. Typically 1 pass is sufficient.

Step 9 – Hit the "**Start Scan**" button. The system will acquire the images and automatically start data analysis.

Step 10 – After scan completion of the 8USCTL123service (or HDCTL 123serv) mode, the tool will automatically advance to scan in 8USCTL456 service (or HDCTL 456serv) mode.

3-2-3 Data Analysis

When the scan completes, the SNR data will be displayed in the Multicoil SNR Tool Control window (see *Figure 7*). The individual elements are assigned a pass/fail value. The results of section 3-2 are automatically recorded as files in /usr/g/service/data. Please refer to this directory for a history of functional tests performed on a specific system.

The individual receiver images and composite image are saved in the system. The signals used for calculate SNR are taken from the composite image (see *Figures 8 and 9*).

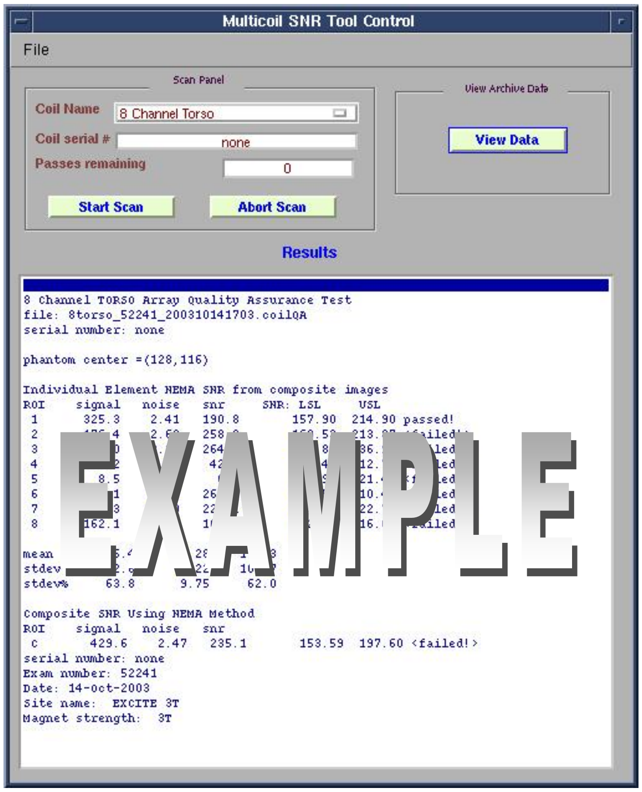

Figure 7: Results displayed in the Multicoil SNR Tool Control window.

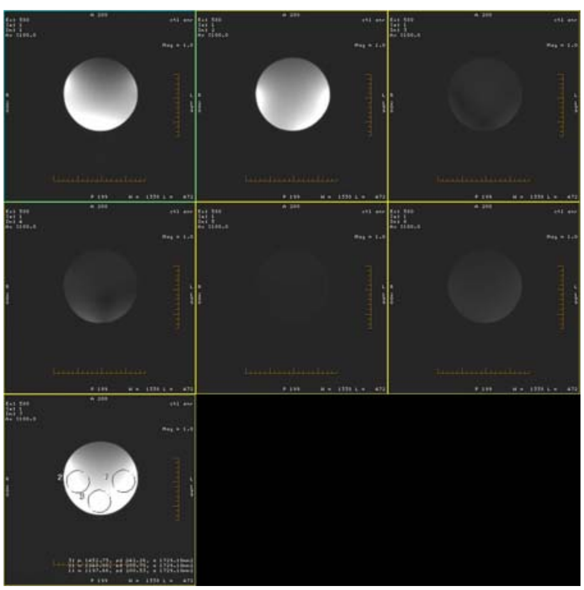

Figure 8: The individual receiver images and composite image of 8USCTL123service (or HDCTL 123serv) mode

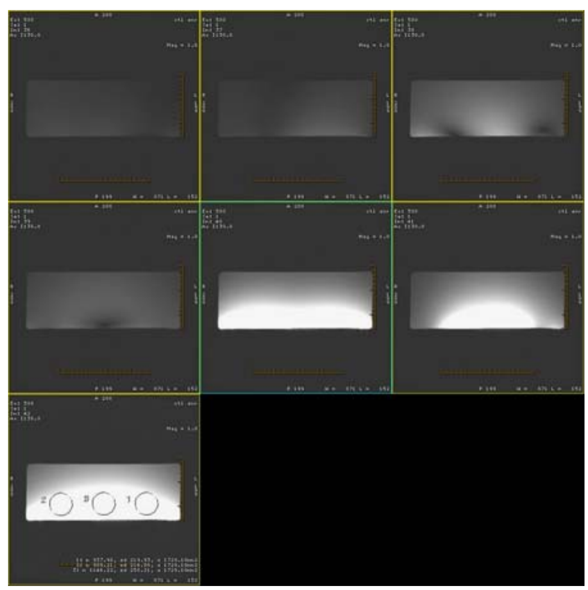

Figure 9: The individual receiver images and composite image of 8USCTL456service (or HDCTL 456serv) mode

To View Archived Data Files:

Step 1 – Click on the "**View Data**" button in the Multicoil SNR Tool Control window (see *Figure 7*). The Multicoil SNR Display Control window opens (see *Figure 10*).

Step 2 – Select the [**FileTab**] in the Multicoil SNR Display Control window.

Step 3 – Turn Expert Mode on.

Step 4 – Select the 8-Channel CTL coil in the coil selection menu.

Step 5 – Click the “**Plot Data**” button.

Step 6 – The Data Plot Control window opens (see *Figure 11*).

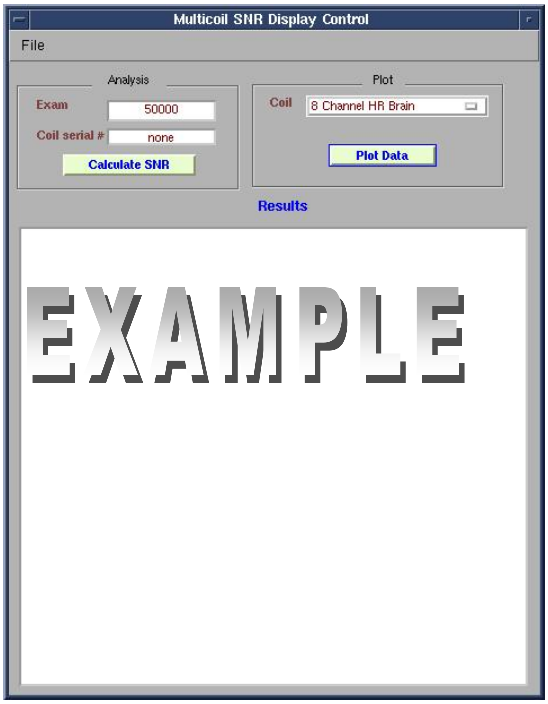

Figure 10: Multicoil SNR Display Control window.

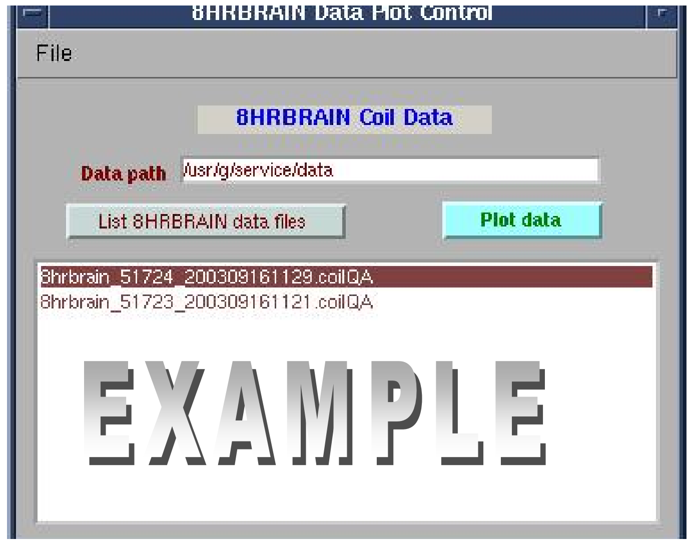

Figure 11: Data Plot Control window.

Step 7 – Select the desired file and click “**Plot Data**”.

Step 8 – A plot window with SNR plots will open (see *Figure 12*).

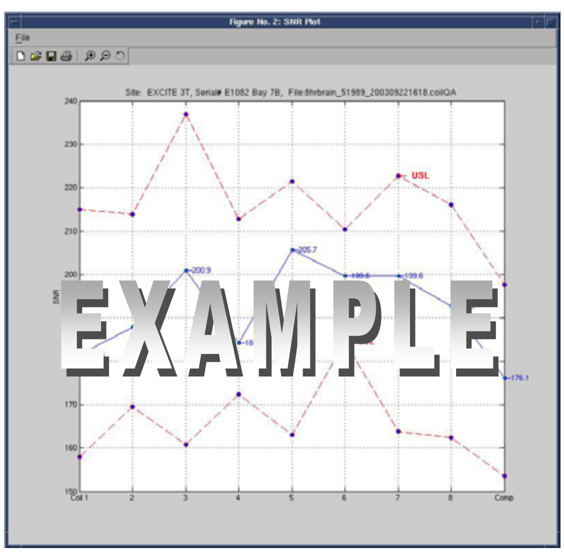

Figure 12: Nominal Individual Element results.

3-2-4 Test Explanation and Results Interpretation

The Premier III Phased Array CTL Spine Coil is a complex phased array. The Quality Assurance tool acquires data looking at the composite image (like the customer sees). Individual element performance is evaluated using the composite image.

Signal images are acquired using a SE sequence with CV saveinter set to 1 to save intermediate images into the image database. The scan is taken twice for a total of 42 images in the 8USCTL123service (or HDCTL 123serv) and 8USCTL456service (or HDCTL 456serv) modes (see *Figures 8 and 9*).

The test employs a filtered version of the NEMA SNR measurement method. The filter takes out adverse effects of subtle ghosting on noise measured in difference.

3-3 External Cable Check

Visual check

Before removing the cable assembly from the coil, carefully check all coaxial connectors and pins on the coil connector. *Figure13* shows the pin layouts of the coil connector. The connector of the CTL coil has 12 pins in Row B, eight pins in Row C and eight coaxial connectors in Row D, If there are any broken, deformed or recessed pins or coaxial connectors, replace the cable assembly.

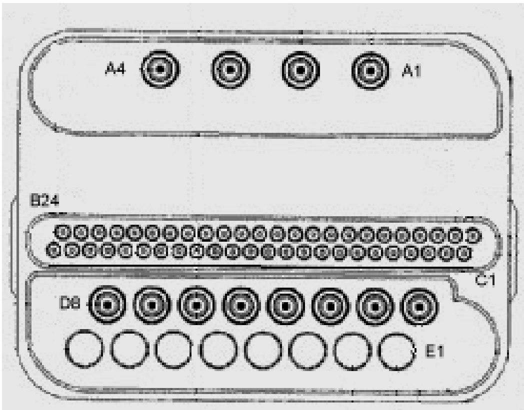

Figure 13: Pin layouts of Hypertronics connector.

Check for continuity of DC lines in the external cable

Step 1 – Remove the cable assembly from the coil as described in Section 5-1, External Cable Replacement.

Step 2 – There are six pins in Row B (see *Figure 13*) connecting to DC wires of the cable. *Table 3-3* shows these pins and corresponding DC wires. Use a Digital Volt Meter (DVM) to check the resistance between the pin in Row B and the corresponding DC wire (see *Figure 14*). The resistance should be 6.0 $\pm$ 0.4Ω. If any resistance reading is not within this range, replace the cable assembly.

PINS AND CONNECTED DC WIRES – TABLE 3-3

Pin in Row B	10	11	12	13	14	15
DC wire color	Black	Brown	Red	Yellow	Green	Blue
DC bias of the system	MC1	MC2	MC3	MC4	MC5	MC6

Step 3 – Check if any pin from B10 to B15 is shorted to the ground. The outside conductor of a coaxial connector in Row D can be used as ground. If any pin is shorted to ground, replace the cable assembly.

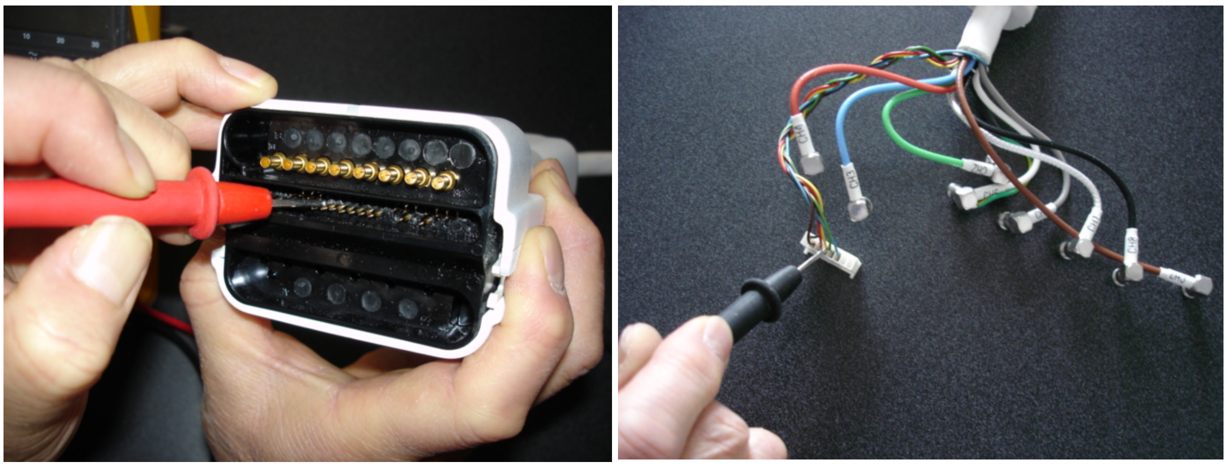

Figure 14: Measure the resistance of the DC line.

Check for continuity of the preamp bias

The Pin 21 in row C is connected with eight RF cables. The system supplies the preamp bias voltage through Pin C21 and RF cables to the coil. Referring to *Figure 15*, find Pin 21 in Row C. Use a DVM to check resistance between Pin C21 and the central pin of the SMB connector at the end of each RF cable (see *Figure 15*). The resistance should be 6.0 ± 0.4Ω. If any resistance reading is not within this range, replace the cable assembly.

Figure 15: Measure the resistance of the preamp bias line.

3-4 PIN Diodes Check

This coil does not contain replaceable PIN diodes. No PIN diodes check is required.

3-5 Troubleshooting Tips

Symptom #1: The system reports a coil fault during prescan or does not recognize the coil connection to the system when selected in the software.

Probable Cause	Suggested Actions	Resolution
The coil connector has become disconnected from the system interface.	Check to make sure the coil connector is fully engaged.	Engage connector and try the scan again.
There is a DC short or open somewhere within the DC path inside the coil.	Perform Section 3-3, External Cable Check.	If there is no cable damage detected, replace the coil.
The output cable has a short or an open.	Perform Section 3-3, External Cable Check, to confirm.	Replace the coil cable assembly.
There is a problem with the DC bias from the system interface	Check to ensure the correct DC bias is being provided to the coil during transmit and receive per GE MRI System Service Manual. Try a similar 8-channel coil to see if the same problem occurs	Correct MRI system problem.
Coil ID issues.	From CSD, select Diagnostics, Manual Tests, SRI Functional Checks, and run the SRI Coil ID test. Compare the coil ID value found to that listed in the coil configuration.	If the value does not match, or no coil ID is found, replace the cable assembly.

Symptom #2: The coil does not pass the system level signal to noise check per Coil Imaging Performance Verification test or exhibits poor image quality on patient scans.

Probable Cause	Suggested Actions	Resolution
One or more of the coil channels has a high noise level.	Look at the uncombined images to determine which channel has the problem. Compare the noise standard deviation measurements between channels and system performance logs; they should be approximately the same.	Replace the coil.
One or more of the coil channels has low signal.	Look at the uncombined images to determine which channel has the problem. Compare signal mean measurements between channels and system performance logs.	Replace the coil.
Excessive ghosting is causing the noise standard deviation measurements to be artificially high.	Window and level the images down to look at the background for signs of ghosting. Try padding the phantom in the coil. Try turning off the cold heads to see if the ghosting is vibration-related.	If padding or changing the phantom position can minimize ghosting, then the problem is caused by excessive vibration elsewhere in the system. If the ghosting is not symptomatic of phantom positioning or padding, then replace the coil.
There is a problem with the DC bias from the system interface.	Check to ensure the correct DC bias is being provided to the coils during transmit and receive per GEMS System Service Manual.	Correct MRI system DC bias problem.

SECTION 4 – MAINTENANCE

4-1 Coil Care

WARNING!

Detach coil connector from the scanner before attempting to clean. Do not reattach after cleaning until the coil has dried completely. Having the coil attached to the system during cleaning or when it is wet may result in electrical shock.

CAUTION

4-1-1 Coil Care

The Premier III Phased Array CTL Spine Coil and patient comfort pads must be cleaned when needed and stored using the following procedure. Wipe with a cloth that has been dampened in a solution of 10 percent bleach and 90 percent tap water, or 30 percent isopropyl alcohol and 70 percent tap water.

Do not spray or pour any cleaning solution directly on the coil!

Store the coil in an air-conditioned scan room or equipment room.

Under no circumstances should the coil be placed into any type of sterilizer.

4-1-2 Storage Requirements

Store the coil in an air-conditioned scan room or equipment room.

To store the coil, a storage space exceeding the coil dimensions of 18.5” W x 44.7” L x 9.3” H is required.

SECTION 5 – REPLACEMENT

Simple removals that are clearly obvious are not described here.

Unless otherwise noted, the steps for re-assembly are simply the reverse order of the steps described for disassembly.

5-1 External Cable Replacement

Step 1 – Remove coil bottom cover by removing all screws around the perimeter and down the center of the base.

Step 2 – Remove the cable clamp that holds down the cable assembly (see *Figure 16*).

Step 3 – Disconnect the SMB plugs and DC plug on the cable end from the interface boards.

Step 4 – Remove the old cable assembly.

Step 5 – Attach the new cable assembly (see *Figure 17*) to the coil and make sure the cable strain relief matches the notch of the former (see *Figure 18*). The side with “do not” label of the cable connector should face up (see *Figure 18*). If it doesn’t face up, rotate the strain relief. Make sure that the cable clamp holds the cable jacket, and then tighten the cable clamp with the screws. Connect all eight SMB plugs and the DC Plug to the corresponding receptacles on the interface board. The receptacles are marked CH1-8 to correspond with the labels on the cables. Don’t route RF cables and DC wires above the bosses shown in *Figure 16*. Otherwise the corresponding bosses on the bottom cover will squeeze the cables and DC wires when the bottom cover is closed.

Step 6 – Reattach the coil bottom cover.

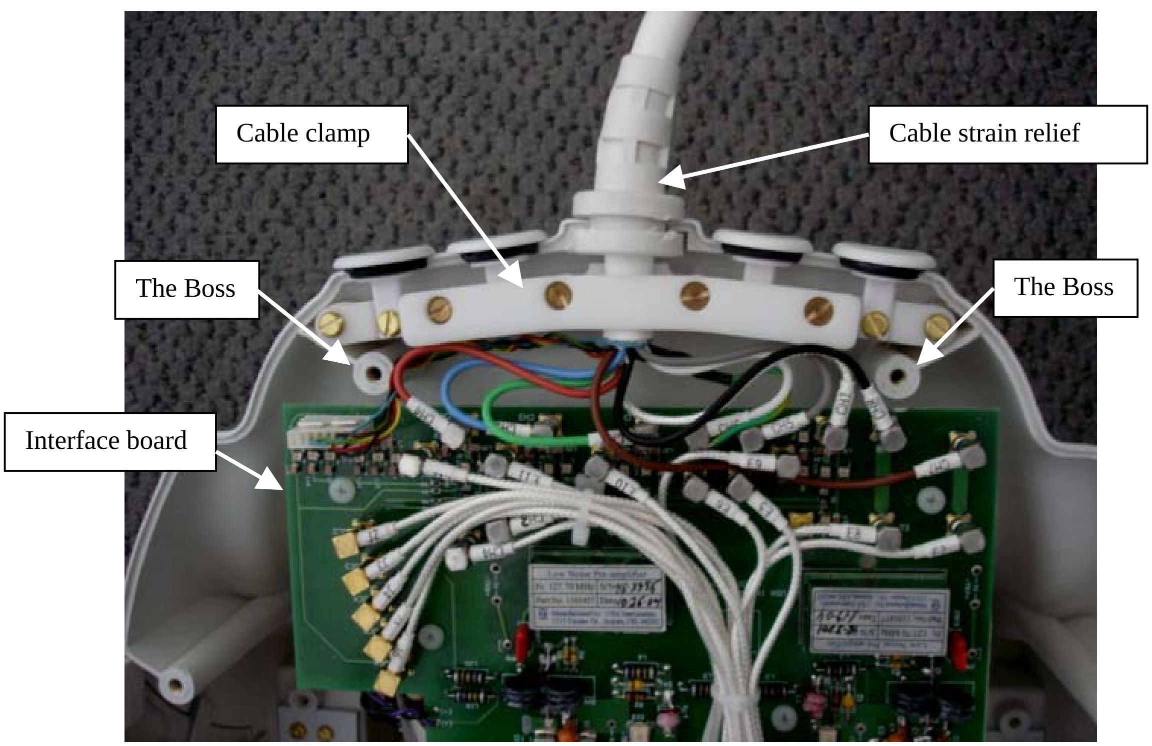

Figure 16: The connection of the cable assembly and the interface board.

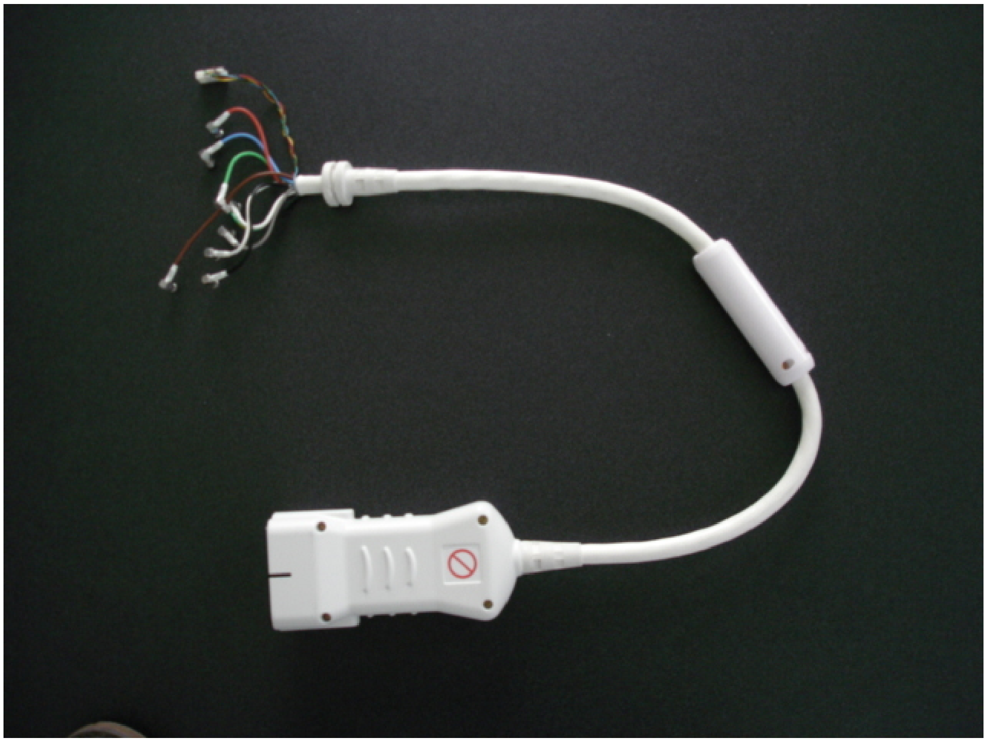

Figure 17: The cable assembly of the CTL coil.

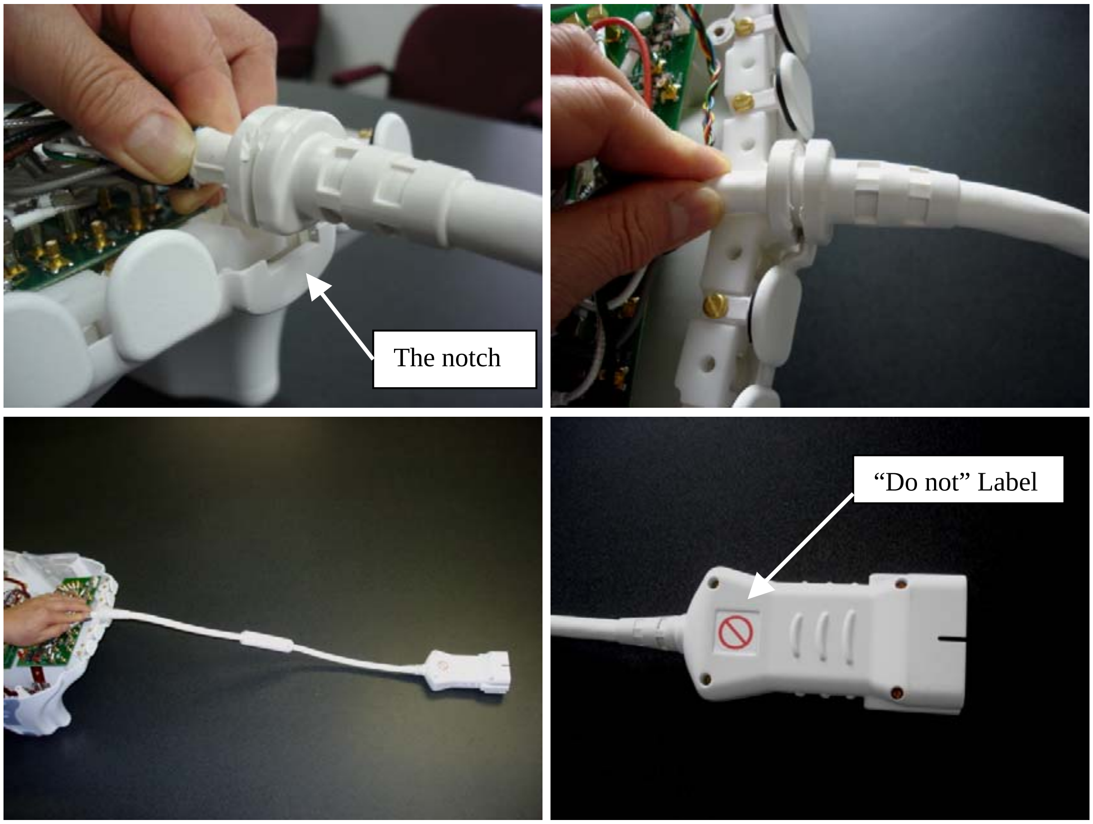

Figure 18: Install the cable assembly

SECTION 6 – RENEWAL PARTS

6-1 Field Replaceable Units

FIELD REPLACEABLE UNITS – TABLE 6-1

Part Name	GE Part #
Premier III Phased Array CTL Spine Coil (GE 3.0T HD CTL Array) w/ Cable	2413107-1
Premier III Phased Array CTL Spine Coil (GE 3.0T HD CTL Array) Cable	2413107-4
3T 8Ch CTL Phantom CT Section	2381682-9
3T 8Ch CTL Phantom TL Section	2381682-10

6-2 Other Replaceable Accessories

OTHER REPLACEABLE ACCESSORIES – TABLE 6-2

Part Name	GE Part #
Patient Comfort Pad	2413107-5
Ear Pad Set	2414538
Disposable Ear Pad Covers	2380634-8

REVISION HISTORY

Date	Rev	ECO	Author	Change Description
01/05	01	N/A	Limin Feng	First Issue
08//20/08	02	2008216	Tricia Luthringer	Table 1-1: Changed p/n's for coil and Op guide. Section 1-1: Revised picture of coil. Section 1-3: changed p/n for Op guide. Section 3-2-2: Changed the Landmark information and step 5. Revised page header. Section 5-1: Revised Step 5 & 6. Revised tables 2-1-1 and 2-1-2.